

CHAPTER 13

HYDRODYNAMICS

The study of fluids in motion

The quantity of liquid flowing through a pipe along a trough, through an orifice, etc., is usually expressed either as the volume flowing per unit time or the mass flowing per unit time. The volume per unit time is termed the volume flow (m^3/s).

The mass flowing per unit time (which is volume flow $\times$ density) is termed the *mass flow* (kg/s).

The velocity at which the liquid flows is the 'length' of liquid which passes a given point in unit time. For instance, if the velocity is 2 m/s it can be thought of as a column of liquid 2 m long passing a point every second, therefore:

$$\text{Volume flow } (\dot{V}) = \text{area} \times \text{velocity}$$

Example. Oil of relative density 0.9 flows at full bore through a pipe 75 mm internal diameter at a velocity of 1.2 m/s. Calculate the mass flow in t/h.

$$\begin{aligned}\text{Volume flow} &= 0.7854 \times 0.075^2 \times 1.2 \\ &= 5.303 \times 10^{-3} \text{ m}^3/\text{s}\end{aligned}$$

$$\begin{aligned}\text{Density of oil of relative density 0.9} \\ &= 0.9 \times 10^3 \text{ kg/m}^3 = 0.9 \text{ t/m}^3\end{aligned}$$

$$\begin{aligned}\text{Mass flow } (\dot{m}) &= 5.303 \times 10^{-3} \times 0.9 \times 3600 \\ &= 17.18 \text{ t/h} \quad \text{Ans.}\end{aligned}$$

FLOW THROUGH VALVES. Referring to Fig. 217, the area of escape is the annular area of the circumferential opening between valve and seat which is, circumference $\times$ lift. The lift beyond which it would cause no restriction is when the circumferential area of lift opening is equal to the cross-sectional area of the bore, thus,

$$\text{Circumference} \times \text{Lift} = \text{Cross-sectional area}$$

$$\pi d \times L = \frac{\pi}{4} d^2$$

$$L = \frac{d}{4}$$

Hence a lift equal to one-quarter of the diameter of the valve allows full bore flow.

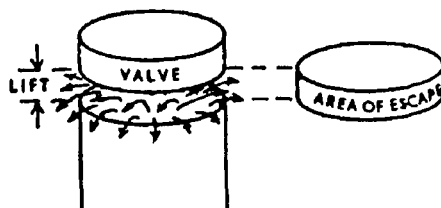


Fig. 217

Example. Calculate the volume flow of water in m^3/min through a 150 mm diameter valve when the velocity of the water is 2.5 m/s and the valve lift is (i) 30 mm, (ii) 45 mm.

Maximum effective lift = $\frac{1}{4} \times 150 = 37.5$ mm

when lift is 30 mm:

$$\begin{aligned} \text{Volume flow} &= \text{area} \times \text{velocity} \\ &= \text{circumference} \times \text{lift} \times \text{velocity} \\ &= \pi \times 0.15 \times 0.03 \times 2.5 \times 60 \\ &= 2.12 \text{ m}^3/\text{min} \quad \text{Ans. (i)} \end{aligned}$$

when lift is equal to or more than quarter diameter:

$$\begin{aligned} \text{Volume flow} &= \text{cross-sect. area} \times \text{velocity} \\ &= 0.7854 \times 0.15^2 \times 2.5 \times 60 \\ &= 2.651 \text{ m}^3/\text{min} \quad \text{Ans. (ii)} \end{aligned}$$

DISCHARGE THROUGH AN ORIFICE

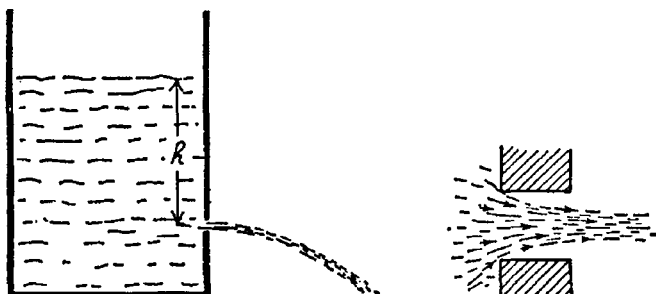


Fig. 218

When water escapes through a hole in the side of a tank (Fig. 218) the potential energy of the water inside the tank due to its head h above the hole is converted into kinetic energy as it flows through the hole, thus,

Kinetic energy gained = Potential energy lost

$$\frac{1}{2}mv^2 = mg \times h$$

$$v = \sqrt{2gh}$$

This is the theoretical velocity of the water jet issuing from the hole and this multiplied by the area of the hole will give the theoretical volume flow.

Due to friction, the actual velocity is a little less, the ratio of the actual velocity to the theoretical velocity is termed the *coefficient of velocity* and represented by C_v hence,

$$\text{Actual velocity} = C_v \sqrt{2gh}$$

Due to eddy currents, the actual area of the water jet is less than the area of the hole, the ratio between the two being termed the *coefficient of reduction of area* and represented by C_A , hence,

$$\text{Area of jet} = C_A \times \text{area of hole.}$$

The actual volume flow is therefore,

$$\text{Volume flow} = C_A \times \text{area of hole} \times C_v \sqrt{2gh}$$

The ratio of the actual quantity discharged to the theoretical quantity is the *coefficient of discharge*, it is represented by C_D and its value is,

$$C_D = C_A \times C_v$$

Example. Water escapes through a hole 20 mm diameter in the side of a tank, the head of water above the hole being 3 m. Taking the coefficient of velocity as 0.97 and the coefficient of reduction of area as 0.64, calculate (i) the velocity of the water jet as it leaves the hole, (ii) the quantity of water escaping in t/h.

$$\begin{aligned} \text{Velocity of water jet} &= 0.97 \times \sqrt{2gh} \\ &= 0.97 \times \sqrt{2 \times 9.81 \times 3} \\ &= 7.442 \text{ m/s} \quad \text{Ans. (i)} \end{aligned}$$

$$\begin{aligned} \text{Area of water jet} &= 0.64 \times \text{area of hole} \\ &= 0.64 \times 0.7854 \times 20^2 \\ &= 201 \text{ mm}^2 = 201 \times 10^{-6} \text{ m}^2 \end{aligned}$$

$$\begin{aligned}
 \text{Volume flow} &= \text{area} \times \text{velocity} \\
 &= 201 \times 10^{-6} \times 7.442 \times 3600 \\
 V &= 5.387 \text{ m}^3/\text{h} \\
 \text{Mass flow} &= \text{vol. flow} \times \text{density} \\
 &= 5.387 \times 1 \\
 \dot{m} &= 5.387 \text{ t/h} \quad \text{Ans. (ii)}
 \end{aligned}$$

Example. In an experiment the water level in a tank was kept constant at 1.25 m above a hole 12 mm diameter in the side of a tank. The jet of water from the hole passed through a ring which was 2.17 m horizontally from the side of the tank and 1 m vertically below the hole. The water discharged into a tank at the rate of 20.84 l/min. From these results calculate the coefficients of velocity, reduction of area and discharge.

$$\begin{aligned}
 \text{Theoretical horizontal velocity of the jet} &= \sqrt{2gh} \\
 &= \sqrt{2 \times 9.81 \times 1.25} = 4.952 \text{ m/s}
 \end{aligned}$$

Let t = time for water to fall from hole to ring, a vertical distance of 1 m.

$$\begin{aligned}
 \text{Initial vertical velocity} &= 0 \\
 \text{Vertical velocity after } t &= 9.81 \times t \text{ m/s} \\
 \text{Average vertical velocity} &= \frac{1}{2} \times 9.81 t \\
 \text{Vertical distance} &= \text{average velocity} \times \text{time} \\
 1 &= \frac{1}{2} \times 9.81 \times t \times t \\
 t &= \sqrt{\frac{2}{9.81}} = 0.4514 \text{ s} \\
 \text{Horizontal distance} &= \text{horizontal velocity} \times \text{time} \\
 \text{Horizontal velocity} &= \frac{2.17}{0.4514} = 4.807 \text{ m/s} \\
 \text{Coeff. of velocity} &= \frac{\text{Actual velocity}}{\text{Theoretical velocity}} \\
 &= \frac{4.807}{4.952} = 0.9705 \\
 \text{Theo. discharge} &= \text{Area of hole} \times \sqrt{2gh} \\
 &= 0.7854 \times 0.012^2 \times 4.952 \\
 &= 5.602 \times 10^{-4} \text{ m}^3/\text{s}
 \end{aligned}$$

$$(1\text{m}^3 = 10^3 \text{ l})$$

$$\begin{aligned}\text{Theo. discharge [l/min]} &= 5.602 \times 10^{-4} \times 10^3 \times 60 \\ &= 33.61 \text{ l/min}\end{aligned}$$

$$\begin{aligned}\text{Coeff. of discharge} &= \frac{\text{Actual discharge}}{\text{Theoretical discharge}} \\ &= \frac{20.84}{33.61} = 0.62\end{aligned}$$

$$\begin{aligned}\text{Coeff. of reduction of area} &= \frac{\text{Coeff. of discharge}}{\text{Coeff. of velocity}} \\ &= \frac{0.62}{0.9705} = 0.6389\end{aligned}$$

$$\left. \begin{array}{l} C_v = 0.9705 \\ C_A = 0.6389 \\ C_D = 0.62 \end{array} \right\} \text{ Ans.}$$

f BERNOULLI'S EQUATION

Bernoulli's theorem states that the total energy contained in a given quantity of water is composed of (i) potential energy, by virtue of its height, (ii) pressure energy, by virtue of its pressure, and (iii) kinetic energy, by virtue of its velocity.

If there is no friction loss and no work done between two points 1 and 2 of a water system, then,

Total energy in water at point 1 = Total energy at point 2

$$\text{Pot.E.}_1 + \text{Press.E.}_1 + \text{K.E.}_1 = \text{Pot.E.}_2 + \text{Press.E.}_2 + \text{K.E.}_2$$

Let m represent the mass of water in kg,

Potential energy is the product of the force of gravity on the mass (*i.e.* its weight) and the height above a given datum level, denoted here by symbol z , thus,

$$\text{Potential energy} = mgz$$

Pressure energy is the product of the pressure and the volume, thus,

$$\text{Potential energy} = pV$$

Kinetic energy is the translational energy of motion, thus,

$$\text{Kinetic energy} = \frac{1}{2}mv^2$$

Inserting these values into the equation:

$$mgz_1 + p_1V + \frac{1}{2}mv_1^2 = mgz_2 + p_2V + \frac{1}{2}mv_2^2$$

Dividing every term by mg , each term will represent its respective energy in J/N weight of the liquid (Nm/N is an equivalent head).

$$z_1 + \frac{p_1 V}{mg} + \frac{v_1^2}{2g} = z_2 + \frac{p_2 V}{mg} + \frac{v_2^2}{2g}$$

mg/V is the specific weight of the liquid (w) N/m³.

$$z_1 + \frac{p_1}{w} + \frac{v_1^2}{2g} = z_2 + \frac{p_2}{w} + \frac{v_2^2}{2g}$$

where z = head of liquid (m)

p = pressure (N/m²)

w = specific weight (N/m³)

v = velocity (m/s)

g = 9.81 (m/s²)

f Example. Part of a vertical fresh water pipe line tapers uniformly from 120 mm diameter at the bottom to 60 mm diameter at the top, the difference in height being 5 m. When the volume flow is 0.0424 m³/s the pressure at the bottom is 160 kN/m², find the pressure at the top.

$$\text{Velocity} = \frac{\text{volume flow}}{\text{area}}$$

$$v_1 = \frac{0.0424}{0.7854 \times 0.12^2} = 3.75 \text{ m/s}$$

$$v_2 = \frac{0.0424}{0.7854 \times 0.06^2} = 15 \text{ m/s}$$

Alternatively, as volume flow is constant,

$$\dot{V} = v_1 \times a_1 = v_2 \times a_2$$

$$v_2 = \frac{3.75 \times 0.7854 \times 0.12^2}{0.7854 \times 0.06^2}$$

$$= 3.75 \times 2^2 = 15 \text{ m/s}$$

Taking bottom as datum level, $z_1 = 0$ and $z_2 = 5$

Let p_2 = pressure at top

$$z_1 + \frac{p_1}{w} + \frac{v_1^2}{2g} = z_2 + \frac{p_2}{w} + \frac{v_2^2}{2g}$$

$$0 + \frac{160 \times 10^3}{10^3 \times 9.81} + \frac{3.75^2}{2 \times 9.81} = 5 + \frac{p_2}{10^3 \times 9.81} + \frac{15^2}{2 \times 9.81}$$

$$160 + \frac{3.75^2}{2} = 5 \times 9.81 + \frac{p_2}{10^3} + \frac{15^2}{2}$$

$$160 + 7.03 = 49.05 + \frac{p_2}{10^3} + 112.5$$

$$\frac{p_2}{10^3} = 160 + 7.03 - 49.05 - 112.5$$

$$p_2 = 5.48 \text{ kN/m}^2 \text{ Ans.}$$

f EFFECT OF FRICTION

In any pipe system, there will be some energy lost by the liquid in overcoming frictional resistance, as the liquid flows through the pipes and valves.

Thus, taking energy loss due to friction into account, between any two points in a system:

$$\begin{array}{ccccc} \text{Initial energy} & - & \text{Friction} & = & \text{Final energy} \\ \text{of liquid} & & \text{losses} & & \text{of liquid} \end{array}$$

$$\text{Initial energy} = \text{Final energy} + \text{Friction losses}$$

Hence, referring to Bernoulli's equation,

$$z_1 + \frac{p_1}{w} + \frac{v_1^2}{2g} = z_2 + \frac{p_2}{w} + \frac{v_2^2}{2g} + \text{Friction losses}$$

Note: To conform with the other terms in the equation the energy loss due to friction must be expressed in J/N weight of the liquid. *i.e.* the equivalent m head.

f Example. Fresh water, density 1000 kg/m³ flows upward through a tapered pipe 6.1 m long 102 mm diameter at the bottom and 204 mm diameter at the top. The pressure at the bottom is 311 kN/m², while the velocity at the top is 9.15 m/s. Determine the velocity at the bottom and the pressure at the top, if the energy loss due to friction is equivalent to 32 m head of water.

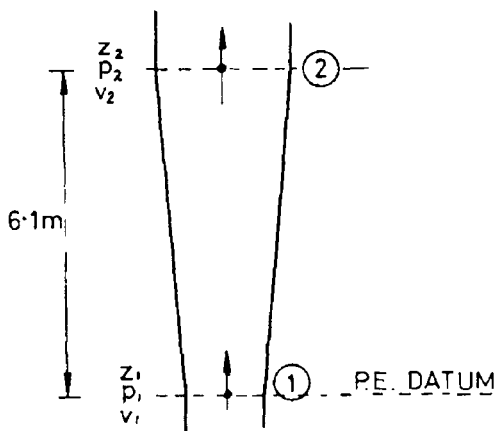


Fig. 219

Denoting conditions at the bottom by suffix 1 and conditions at the top by suffix 2:

Since volumetric flow rate is constant,

$$\text{area}_1 \times \text{velocity}_1 = \text{area}_2 \times \text{velocity}_2$$

$$\frac{\pi}{4} \times d_1^2 \times v_1 = \frac{\pi}{4} \times d_2^2 \times v_2$$

$$0.102^2 \times v_1 = 0.204^2 \times 9.15$$

$$v_1 = 36.6 \text{ m/s. Ans.}$$

From Bernoulli,

$$z_1 + \frac{p_1}{w} + \frac{v_1^2}{2g} = z_2 + \frac{p_2}{w} + \frac{v_2^2}{2g} + \text{Friction loss}$$

$$0 + \frac{311 \times 10^3}{10^3 \times 9.81} + \frac{36.6^2}{2 \times 9.81} = 6.1 + \frac{p_2}{10^3 \times 9.81} + \frac{9.15^2}{2 \times 9.81} + 32$$

$$\therefore 31.7 + 68.27 = 6.1 + \frac{p_2}{10^3 \times 9.81} + 4.27 + 32$$

$$\therefore p_2 = 57.6 \times 10^3 \times 9.81$$

$$\text{Pressure at the top} = 565 \text{ kN/m}^2. \text{ Ans.}$$

f VENTURI METER

A venturi meter is a device for measuring the rate of flow of liquid through a pipeline. It consists of a short length of pipe which tapers from full bore at each end to a smaller bore (termed the throat) at the middle, as illustrated in Fig. 220.

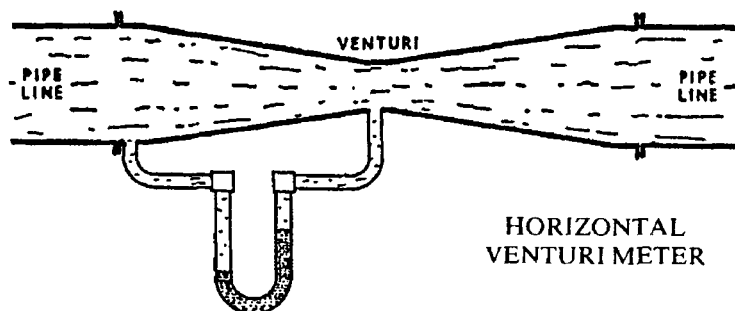


Fig. 220

The areas at entrance and throat are known, and the pressures at these two points, or difference in pressure, is measured, then Bernoulli's equation can be applied to find the velocity of the liquid and hence the quantity flowing.

However, it is usual to connect a venturi meter horizontally in a horizontal part of the pipeline, and this simplifies calculations as there is no difference in head between the two points, that is, $z_1 = z_2$ and cancels. Let the entrance be denoted by point number 1 and the throat by point number 2, for a *horizontal* meter:

$$\text{press. energy}_1 + \text{kinetic energy}_1 = \text{press. energy}_2 + \text{kinetic energy}_2$$

$$\text{press. energy}_1 - \text{press. energy}_2 = \text{kinetic energy}_2 - \text{kinetic energy}_1$$

which is, in effect,

$$\text{loss of pressure energy} = \text{gain in kinetic energy}$$

$$p_1 V - p_2 V = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2$$

$$V (p_1 - p_2) = \frac{1}{2} m (v_2^2 - v_1^2)$$

but, mass m ÷ volume V = density ρ , therefore, dividing throughout by V ,

$$p_1 - p_2 = \frac{1}{2} \rho (v_2^2 - v_1^2)$$

From the above equation, the velocity (v_1) of the liquid flowing in the pipe may be calculated.

$$\text{Flow rate} = \text{cross sect. area of pipe} \times \text{velocity in pipe}$$

f METER COEFFICIENT

Taking frictional resistance into account, as the liquid flows through the meter, it is found that the loss of pressure energy is *not* equal to the gain in kinetic energy. Some of the pressure energy loss is due to work done against friction. As a result, the calculated flow rate must be corrected by a factor called a *meter coefficient* or *discharge coefficient*. This meter coefficient is determined by comparing the calculated or *theoretical* flow rate and the *actual* flow rate through the meter.

Actual flow rate = Theoretical flow rate $\times$ meter coefficient.

f Example. The diameters at entrance and throat of a horizontal venturi meter are 450 mm and 225 mm respectively, and the difference in pressure between these two points is equivalent to 381 mm head of water. Calculate the mass flow of fresh water through the meter ($\rho = 1000 \text{ kg/m}^3$).

Denoting conditions at entrance by suffix 1, and at throat by 2, since the volume flow rate at any point is constant, then,

$$\text{area}_1 \times \text{velocity}_1 = \text{area}_2 \times \text{velocity}_2$$

$$\begin{aligned} v_2 &= \frac{a_1 \times v_1}{a_2} \\ &= \frac{0.7854 \times 0.45^2}{0.7854 \times 0.225^2} \times v_1 \\ v_2 &= 4v_1 \end{aligned}$$

Difference in pressure = 381 mm water

$$\begin{aligned} p_1 - p_2 &= \rho gh \\ &= 10^3 \times 9.81 \times 0.381 \\ &= 3738 \text{ N/m}^2 \end{aligned}$$

$$p_1 - p_2 = \frac{1}{2} \rho (v_2^2 - v_1^2)$$

Substituting $p_1 - p_2 = 3738 \text{ N/m}^2$

$$\begin{aligned} v_2 &= 4v_1 \\ 3738 &= \frac{1}{2} \times 10^3 \times \{(4v_1)^2 - v_1^2\} \\ 3738 &= \frac{1}{2} \times 10^3 \times 15v_1^2 \end{aligned}$$

$$\begin{aligned} v_1 &= \sqrt{\frac{2 \times 3738}{10^3 \times 15}} \\ &= 0.706 \text{ m/s} \end{aligned}$$

Volume flow = area $\times$ velocity

$$\dot{V} = 0.7854 \times 0.45^2 \times 0.706$$

Mass flow = volume flow $\times$ density

$$= 0.7854 \times 0.45^2 \times 0.706 \times 10^3$$

$$\dot{m} = 112.3 \text{ kg/s} \text{ Ans.}$$

f Example. A pipe 300 mm diameter contains a venturi meter with a throat diameter of 100 mm. The difference in pressure head between inlet and throat is 250 mm of mercury, measured on a U-tube gauge containing water and mercury. The meter coefficient is 0.95. Calculate the discharge rate through the pipe, in m³/s. (Density of water is 1000 kg/m³.)

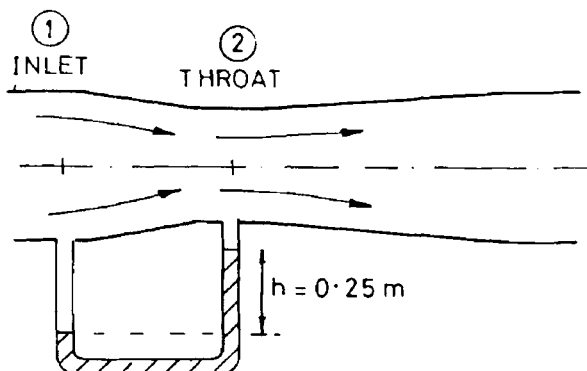


Fig. 221

$$p_1 - p_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) \dots \dots \dots (i)$$

For a differential manometer, containing water and mercury,

$$\begin{aligned} p_1 - p_2 &= h \times 9.81 \times 12.6 \times 10^3 \\ &= 0.25 \times 9.81 \times 12.6 \times 10^3 \text{ N/m}^2 \\ &= 30.9 \text{ kN/m}^2 \end{aligned}$$

Flow rate volume ($\dot{V}$) through the meter is constant,

$$a_1 v_1 = a_2 v_2$$

$$\frac{\pi}{4} \times 0.3^2 \times v_1 = \frac{\pi}{4} \times 0.1^2 \times v_2$$

$$v_2 = \left(\frac{0.3}{0.1} \right)^2 \times v_1$$

$$v_2 = 9v_1$$

Substituting in equation (i)

$$30.9 \times 10^3 = \frac{1}{2} \times 1000 [(9v_1)^2 - v_1^2]$$

$$30.9 \times 10^3 = 500 [81v_1^2 - v_1^2]$$

$$30.9 \times 10^3 = 500 \times 80v_1^2$$

$$v_1 = 0.879 \text{ m/s}$$

$$\text{Volume flow rate} = \frac{\text{theoretical flow rate}}{\text{meter coefficient}}$$

$$\dot{V} = \frac{\pi}{4} \times 0.3^2 \times 0.879 \times 0.95$$

$$= 0.059 \text{ m}^3/\text{s. Ans.}$$

f FRICTION IN PIPES

The laws of friction relating to the flow of liquid through pipes are as follows:-

- Fluid friction (i) is proportional to the interior wetted surface area of the pipe.
- (ii) is proportional to the (velocity)^{*n*} of flow where *n* is approximately, and usually taken as 2,
- (iii) depends upon the roughness of the internal surface of the pipe,
- (iv) is inversely proportional to the cross-sectional area of the pipe bore,
- (v) is independent of the intensity of fluid pressure.

Thus, taking water flowing full bore at a velocity of *v* through a pipe of internal diameter *d* and length *l*,

$$\text{Friction} \propto \frac{\text{wetted surface area} \times \text{velocity}^2}{\text{cross-sectional area}}$$

$$\text{Friction} \propto \frac{\pi d l \times v^2}{\frac{\pi}{4} d^2}$$

$$\text{Friction} \propto \frac{4 l v^2}{d}$$

$$\text{Loss due to friction} = a \text{ constant} \times \frac{4 l v^2}{d}$$

It is usual to state the effect of friction as a factor of $\frac{v^2}{2g}$, that

is expressing it as a loss of head, representing this factor by f ,

$$\text{Loss of head due to friction } (h_f) = \frac{4fl}{d} \times \frac{v^2}{2g}$$

f Example. Fresh water flows at the rate of $1.88 \text{ m}^3/\text{h}$ through a pipe 19 mm internal diameter and 6 m long. Taking the friction factor as 0.01, find the loss due to friction expressed in units of (i) head, (ii) pressure.

$$\dot{V} = av$$

$$\begin{aligned}\text{Velocity} &= \frac{1.88}{3600 \times 0.7854 \times 0.019^2} \\ &= 1.842 \text{ m/s}\end{aligned}$$

$$\text{Loss of head due to friction} = \frac{4fl}{d} \times \frac{v^2}{2g}$$

$$\begin{aligned}h_f &= \frac{4 \times 0.01 \times 6 \times 1.842^2}{0.019 \times 2 \times 9.81} \\ &= 2.184 \text{ m} \quad \text{Ans. (i)}\end{aligned}$$

$$\begin{aligned}p &= \rho gh \\ &= 10^3 \times 9.81 \times 2.184 \\ &= 2.143 \times 10^4 \text{ N/m}^2 \\ &= 21.43 \text{ kN/m}^2 \text{ or } 0.2143 \text{ bar} \\ &\quad \text{Ans. (ii)}\end{aligned}$$

f IMPACT OF WATER JET

ON A STATIONARY PERPENDICULAR PLATE

Consider a jet of water of A cross-sectional area moving with a velocity of v striking a stationary flat plate which is perpendicular to the direction of the water jet, as in Fig. 222

If ρ is the density of the water:

$$\begin{aligned}\text{Mass of water striking plate every second} &= \text{area} \times \text{velocity} \times \text{density} \\ &= Av\rho\end{aligned}$$

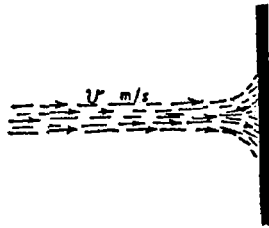


Fig. 222

Assuming that the water has no velocity after striking the plate, its change of velocity is $v - 0 = v$.

Change of momentum = mass $\times$ change of velocity

$$\begin{aligned}\text{Change of momentum / s} &= \text{mass / s} \times \text{change of velocity} \\ &= Av\rho \times v \\ &= A\rho v^2\end{aligned}$$

Change of momentum per second is equal to the force required to cause that change, hence the force applied by the plate on the jet of water is,

$$\text{Force} = A\rho v^2$$

and in accordance with Newton's third law of motion this is also the force applied by the jet on the plate.

ON A MOVING PERPENDICULAR PLATE. In the case of a jet striking a flat plate which is perpendicular to the jet and moving at u m/s in the same direction of the jet:

$$\text{Relative velocity of jet to plate} = (v - u)$$

$$\begin{aligned}\text{Mass of water striking plate every second} \\ &= (v - u) A\rho\end{aligned}$$

Assuming that the water does not splash off the plate with any velocity,

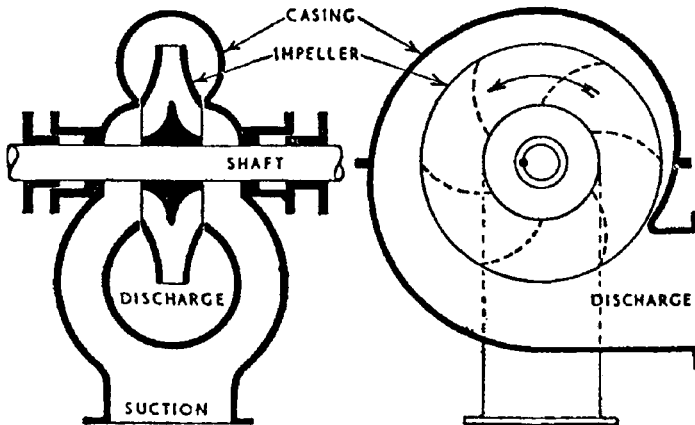
$$\begin{aligned}\text{Change of velocity} &= (v - u) \\ \text{Force of jet} &= \text{Change of momentum / s} \\ &= \text{mass / s} \times \text{change of velocity} \\ &= (v - u) \times A \times \rho \times (v - u) \\ &= A\rho (v - u)^2\end{aligned}$$

In this, flat plates perpendicular to the jet were taken as examples and the assumption made that the water trickled down

the plate after striking, without any splashing or rebound of the water. If curved plates were used such as turbine blades, the water would leave the blade with a certain amount of velocity in a direction depending upon the exit angle of the blade. The change of velocity of the water jet would then be calculated as for any other body subjected to a change of velocity due to combined change of speed and direction, from which the change of momentum per second and therefore the force of the jet can be found.

f CENTRIFUGAL PUMP

This is a rotary pump which works on the principle of centrifugal force. It consists of a rotating impeller within a stationary casing. The impeller is a hollow disc wheel with internal curved vanes, mounted on a shaft which is driven by an electric motor. Openings in the sides of the impeller near the shaft communicate with the suction branch, liquid (oil, water, etc.) enters the rotating impeller through these ports and, due to the circular motion given to the liquid, it is thrown by centrifugal force to the open periphery of the impeller where it is discharged and enters the space between the outer circumference of the impeller and the casing and directed to the outlet branch, see Fig. 223.



CENTRIFUGAL PUMP

Fig. 223

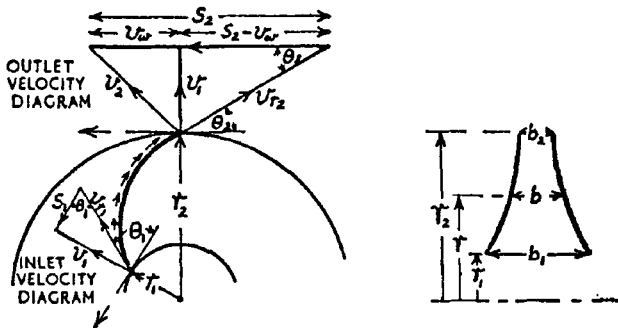


Fig. 224

Referring to the inlet velocity diagram at entrance to the impeller, Fig. 224, it is assumed that the liquid at entrance to the vanes has radial velocity only.

Let s_1 = linear velocity of the vanes at entrance

v_1 = radial velocity of the liquid

v_{r1} = relative velocity of the liquid to the vanes

If the liquid is to enter the vanes without shock, the slope of the vanes at entrance should be the same as the direction of the relative velocity, that is, at θ_1 to the tangent of the inner circle, hence:

$$\tan \theta = \frac{v_1}{s_1} \dots \dots \dots (i)$$

Referring to the outlet velocity diagram at exit from the impeller, after flowing over the surface of the curved vanes, the liquid leaves at a relative velocity of v_{r2} ,

Let s_2 = linear velocity of the vanes at exit

v_2 = absolute velocity of the liquid leaving the vanes

v_w = velocity of whirl of the liquid

v_1 = radial velocity of the liquid which is assumed to be the same at exit as at entrance, and constant on its way through the impeller.

v_1 and v_w are the rectangular components of v_2 . Representing the exit angle of the vanes by θ_2 then:

$$\tan \theta_2 = \frac{v_1}{s_2 - v_w} \dots \dots \dots (ii)$$

The volume flow of the liquid through the impeller is the product of the effective area (circumferential cross-section) of the impeller and the radial velocity of the liquid. Since the volume flow and radial velocity are constant throughout the impeller, then the area of section is constant.

Let A = area of circumferential section of the impeller at radius r where the breadth is b .

$$\text{then } A = 2\pi r b$$

$$\text{Area being constant, } r b = r_1 b_1 = r_2 b_2$$

If the liquid enters the vanes without shock, the tangential velocity of the liquid at entrance is nil. At exit the tangential velocity of the liquid is v_w therefore, if $\dot{m}$ = mass flow per second, the tangential momentum given to water per second is $\dot{m} v_w$.

Since change of momentum per second = force, then the tangential force given to the water is $\dot{m} v_w$.

$$\text{Work /s} = \text{force} \times \text{velocity}$$

$$\text{Work /s imparted to the liquid} = \dot{m} v_w s_2$$

$$\text{Also, Work /s} = \text{force to lift against gravity} \times \text{height}$$

$$= \dot{m} g \times h$$

$$\dot{m} g h = \dot{m} v_w s_2$$

Theoretical head given to water:

$$h = \frac{\dot{m} v_w s_2}{\dot{m} g}$$

$$h = \frac{v_w s_2}{g}$$

f MANOMETRIC HEAD

The manometric head is the *effective* or actual energy imparted to the liquid by a pump.

Liquid enters the system at ①, the suction inlet, and is discharged at ②, the discharge pipe outlet. (see Fig. 225)

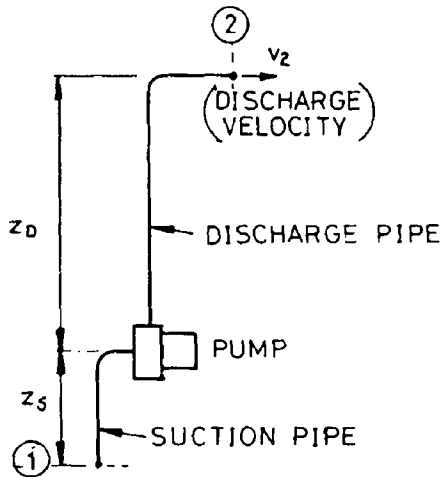


Fig. 225

Manometric head = suction head + discharge head + velocity energy at discharge + friction losses.

$$h_m = z_s + z_d + \frac{v_2^2}{2g} + \text{friction losses}$$

f PUMP POWER

Taking the total energy head into account:

Pump output power = effective work done per second
 = weight of liquid flow / s $\times$ effective head

$$\text{i.e. Pump output power} = \dot{m} \times g \times h_m \quad \text{W}$$

where $\dot{m}$ = mass flow rate of liquid

$$g = 9.81$$

h_m = manometric head

f MANOMETRIC EFFICIENCY

$$\begin{aligned}
 \text{Manometric efficiency} &= \frac{\text{effective head imparted to liquid}}{\text{theoretical head imparted by pump}} \\
 &= \frac{h_m}{\frac{v_w s_2}{g}} \\
 &= \frac{g h_m}{v_w s_2}
 \end{aligned}$$

f Example. The outer and inner diameters of the impeller of a centrifugal pump are 450 and 150 mm diameter respectively, and the width of the impeller at exit is 40 mm. When running at 500 rev/min it discharges 6.8 m³/min against an effective head of 8 m. Calculate the width of the impeller at inlet and, if the water enters the vanes without shock, calculate the angle of the vanes at inlet. The angle of the vanes at exit is 45°, calculate the theoretical head and the manometric efficiency.

Referring to Fig. 224,

$$\begin{aligned}
 r_1 \times b_1 &= r_2 \times b_2 \\
 75 \times b_1 &= 225 \times 40 \\
 b_1 &= 120 \text{ mm Ans. (i)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Radial velocity} &= \frac{\text{volume flow}}{\text{area of flow}} \\
 &= \frac{6.8}{60 \times \pi \times 0.45 \times 0.04} \\
 v_1 &= 2.004 \text{ m/s}
 \end{aligned}$$

Linear velocity of vanes at entrance,

$$\begin{aligned}
 s_1 &= \frac{\pi \times 0.15 \times 500}{60} \\
 &= 3.927 \text{ m/s} \\
 \tan \theta_1 &= \frac{v_1}{s_1} = \frac{2.004}{3.927} = 0.5101
 \end{aligned}$$

Angle at entrance $\theta_1 = 27^\circ 2'$ Ans. (ii)

Linear velocity of vanes at exit,

$$s_2 = \frac{\pi \times 0.45 \times 500}{60}$$

$$= 3 \times s_1 = 11.78 \text{ m/s}$$

$$\frac{v_1}{s_2 - v_w} = \tan \theta_2$$

$$s_2 - v_w = \frac{v_1}{\tan \theta_2}$$

$$= \frac{2.004}{\tan 45^\circ} = 2.004 \text{ m/s}$$

$$v_w = s_2 - (s_2 - v_w)$$

$$= 11.78 - 2.004 = 9.776 \text{ m/s}$$

$$\text{Theoretical head} = \frac{v_w s_2}{g}$$

$$= \frac{9.776 \times 11.78}{9.81}$$

$$= 11.74 \text{ m Ans. (iii)}$$

$$\text{Manometric efficiency} = \frac{\text{effective head}}{\text{theoretical head}}$$

$$= \frac{8}{11.74} = 0.6816 \text{ Ans. (iv)}$$

f ORIFICE CONTROL

Orifice plates with specific characteristics are often inserted into fluid flow lines to control and temperate the mass flow design performance. Questions usually require the solution and manipulation of given formulae. A worked example illustrates the practice as follows:

f Example. The flow of air to a compressor is given by the equation,

$$m \text{ (kg/s)} = C \sqrt{h}$$

$$\text{where } C = 0.816$$

h = the differential pressure head (mm of water)
across an orifice in the pipeline.

(a) The air flow is regulated within the limits of $\pm 5\%$ of the normal flow rate of 10 kg/s.

Calculate the values of h for maximum, normal and minimum flow conditions.

(b) The regulation of the air flow is performed by a pneumatic control system where the control signals are 100 kN/m^2 and 30 kN/m^2 at maximum and minimum flow conditions respectively. The equation for the signal is:

$$S = K_1 + K_2 h$$

where S is the control signal pressure (kN/m^2)

(i) Determine the value of the constants K_1 and K_2 .

(ii) Calculate the control signal pressure when the air flow rate is normal.

$$\text{Maximum air flow rate} = 1.05 \times 10 = 10.5 \text{ kg/s}$$

$$\text{Minimum air flow rate} = 0.95 \times 10 = 9.5 \text{ kg/s}$$

$$\dot{m} = C \sqrt{h}$$

$$h = \left(\frac{\dot{m}}{C} \right)^2$$

$$\left. \begin{array}{l} \text{At max. flow } h = \left(\frac{10.5}{0.816} \right)^2 h = 165.6 \text{ mm} \\ \text{At normal flow } h = \left(\frac{10}{0.816} \right)^2 h = 150.2 \text{ mm} \\ \text{At min. flow } h = \left(\frac{9.5}{0.816} \right)^2 h = 135.5 \text{ mm} \end{array} \right\} \text{Ans. (a)}$$

$$S = K_1 + K_2 h$$

$$\therefore \text{At max. flow: } 100 = K_1 + (K_2 \times 165.6)$$

$$\text{At min. flow: } 30 = K_1 + (K_2 \times 135.5)$$

$$70 = 30.1 K_2$$

$$K_2 = 2.326$$

$$30 = K_1 + (2.326 \times 135.5)$$

$$K_1 = -285.2 \text{ and } K_2 = 2.326 \quad \text{Ans. (b) (i)}$$

$$\text{At normal flow: } S = -285.2 + (2.326 \times 150.2)$$

$$S = 64.16 \text{ kN/m}^2 \quad \text{Ans. (b) (ii)}$$

TEST EXAMPLES 13

1. A pump discharges sea water through a 150 mm diameter pipe at a velocity of 1.75 m/s to a height of 9 m. If the efficiency of the pump is 72%, find the input power. Take the density of sea water as 1025 kg/m³.

2. The diameter of a feed check valve is 60 mm. Find the volume flow in m³/h when the lift of the valve is 10 mm and the velocity of the water is 1.8 m/s.

3. A fire main contains sea water (density 1025 kg/m³) which flows with a velocity of 6 m/s. The main supplies three branches in which water flows at 8 m/s. Branches are 60 mm diameter supplying nozzles of 12 mm diameter. Calculate:

(a) the necessary diameter of the fire main;

(b) the mass flow rate;

(c) the velocity of discharge.

4. Derive a formula for the mass flow of liquid through a valve in t/h , using the symbols, d = diameter in mm, l = lift in mm, v = velocity of water in m/s, ρ = density of liquid in g/ml, and assuming one-sixth of the area of opening is blocked by the wings of the valve.

Using the derived formula, find the mass flow (t/h) of water through a 76 mm diameter valve when the lift is 13 mm, velocity of the water 3 m/s, and the density of the water 0.997 g/ml.

5. A rivet falls out of the side of a tank and water escapes through the rivet hole which is 19 mm diameter. If the head of water in the tank above the hole is 2.5 m, calculate (i) the velocity of the water as it issues from the hole, (ii) the litres escaping every second. Take the coefficient of velocity as 0.97 and coefficient of reduction of area as 0.64.

6. Water flows through a 10 mm diameter orifice in the side of a tank which rests on a horizontal ground. The hole is 0.6 m above ground level and the head of water above the hole is 1.5 m. The jet strikes the ground at a horizontal distance of 1.84 m from the side of the tank and the rate of discharge is 15.85 l/min. Find the coefficients of velocity, reduction of area, and discharge.

7. Fresh water flows through a short smooth horizontal pipe of tapered bore. At the large end the velocity of the water is 1.5 m/s.

At the small end the pressure is 14 kN/m^2 less, calculate the velocity at this end.

f8. Water flows through a smooth bore horizontal venturi meter which is 375 mm diameter at entrance and 125 mm diameter at the throat. If the difference in pressure between these two points is equivalent to 457 mm head of water, calculate the mass flow in kg/s.

f9. A vertical pipe 200 mm diameter at the top and 100 mm diameter at the bottom is 6 m long. Sea water (density 1025 kg/m^3) enters at the bottom at 9 m/s where the pressure is 300 kN/m^2 and flows upwards. Head loss due to friction is given by:

$$hf = 0.25 (v_1^2 - v_2^2) / 9 \text{ m}$$

where v_1 is the velocity at inlet and v_2 is the velocity at exit. Determine the water pressure at the top of the pipe.

f10. (a) Identify the three forms of energy that may be possessed by a liquid in motion stating energy forms in J/N units.

(b) Water flows upward through a vertical venturi meter, 5 cm diameter inlet, 2.5 cm diameter throat. The throat is 15 cm above the inlet and differential pressure between inlet and throat is 250 mm of water. Calculate the mass flow rate of water (density 1000 kg/m^3) through the meter whose coefficient is 0.95.

f11. A pipe 20 cm diameter and 100 m long supplies $0.1 \text{ m}^3/\text{s}$. Determine the head lost in friction over the pipe length and the slope of the hydraulic gradient. Assume $f = 0.005$.

f12. A horizontal jet of fresh water 40 mm diameter impinges on a stationary flat plate which is perpendicular to the jet, and the direct force to prevent the plate moving is measured to be 225 N. Assuming that there is no splash back of the water, find the initial velocity of the jet.

f13. The outer and inner diameters of the impeller of a centrifugal pump are 600 and 300 mm respectively, and the width at the impeller entrance is 120 mm. Calculate the width of the impeller at exit and at mid-radius so that the radial velocity of the water through the impeller is constant. Calculate also the angle of the vanes at entrance so that the water enters without shock when the pump is running at 420 rev/min and the radial velocity is 1.65 m/s .

f14. A pump draws sea water from 3 m below its own level and discharges to a height of 76 m above its level. The flow rate is

108 m³/h and the diameter of the discharge pipe is 130 mm. If the friction head loss in the pipes is 13.7 m and the pump efficiency is 65%, calculate the power required to drive the pump. (Density of sea = 1025 kg/m³.)

f15. (a) The flow rate of air through an orifice is given by:

$$\dot{V} = A \times C_D \times \sqrt{2gh}$$

An orifice plate is used to control the air flow to a ventilation duct. When the orifice plate diameter is 350 mm the air flow rate is 5100 m³/h. C_D is 0.6. Determine the differential pressure across the orifice plate in mm of water.

(b) The flow is to be reduced to one third of its original value. Calculate the diameter of the orifice required assuming the differential pressure remains constant. $C_D = 0.58$.

Density: air = 1.22 kg/m³, water = 1000 kg/m³.